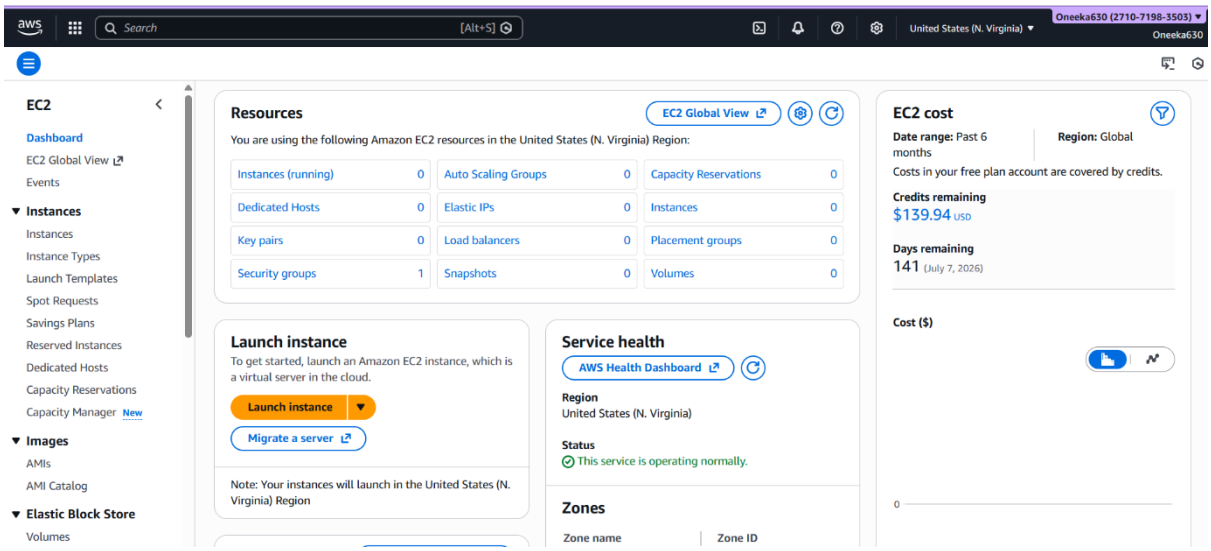
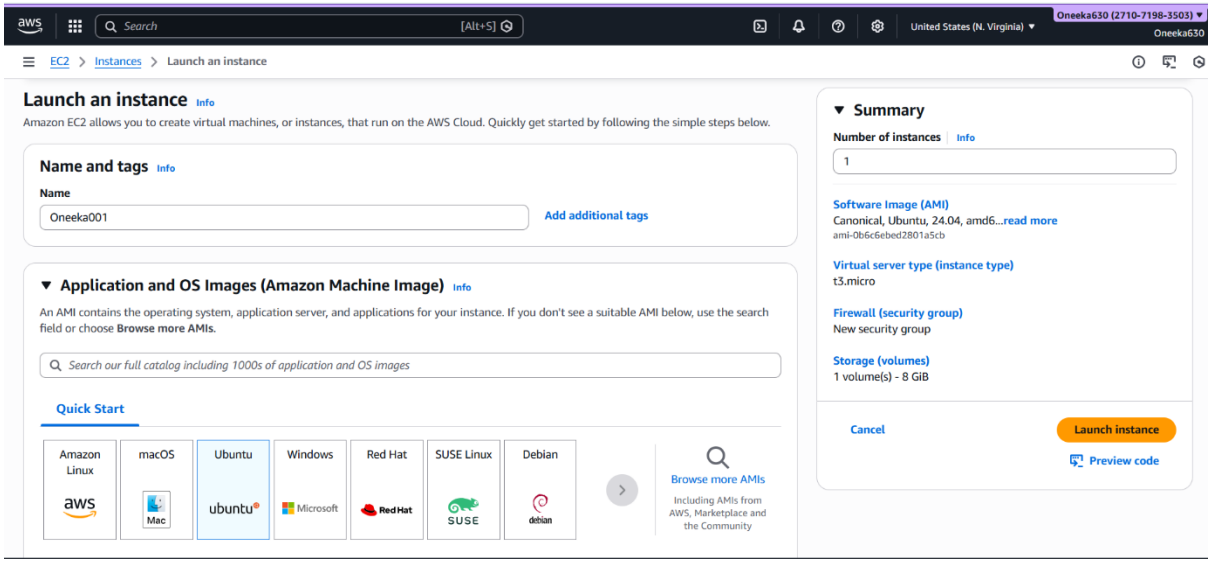
# Assignment No. 9

***Problem Definition: -*** Deploy a project from GitHub to EC2.

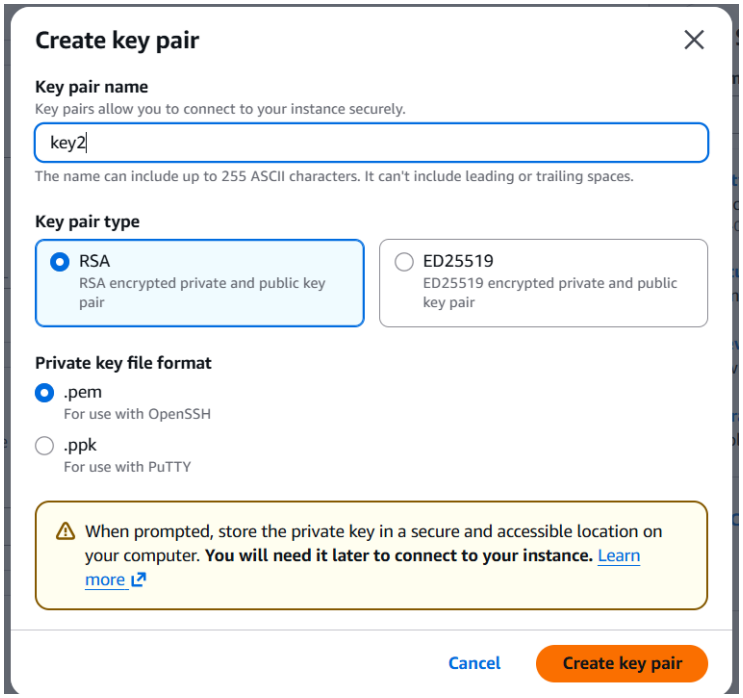
1)Sign up for an **AWS** account, search for '**EC2**' then click on it. -> Click on **'Launch instance'.**

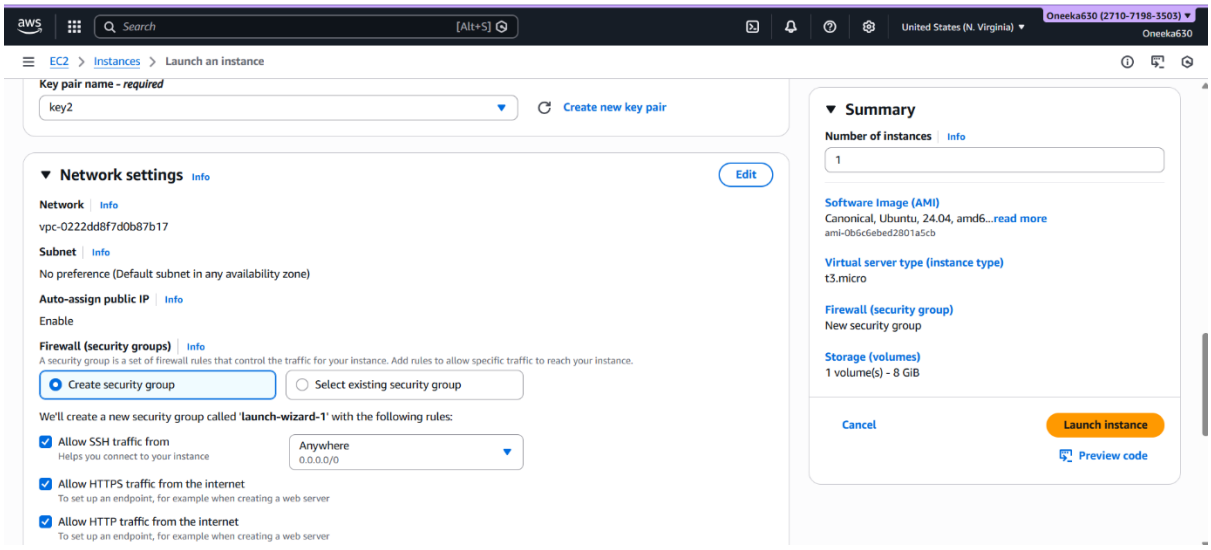



2) Fill up the required details-> '**Name**' and under '**Application and OS Images**' select **'ubuntu'.**

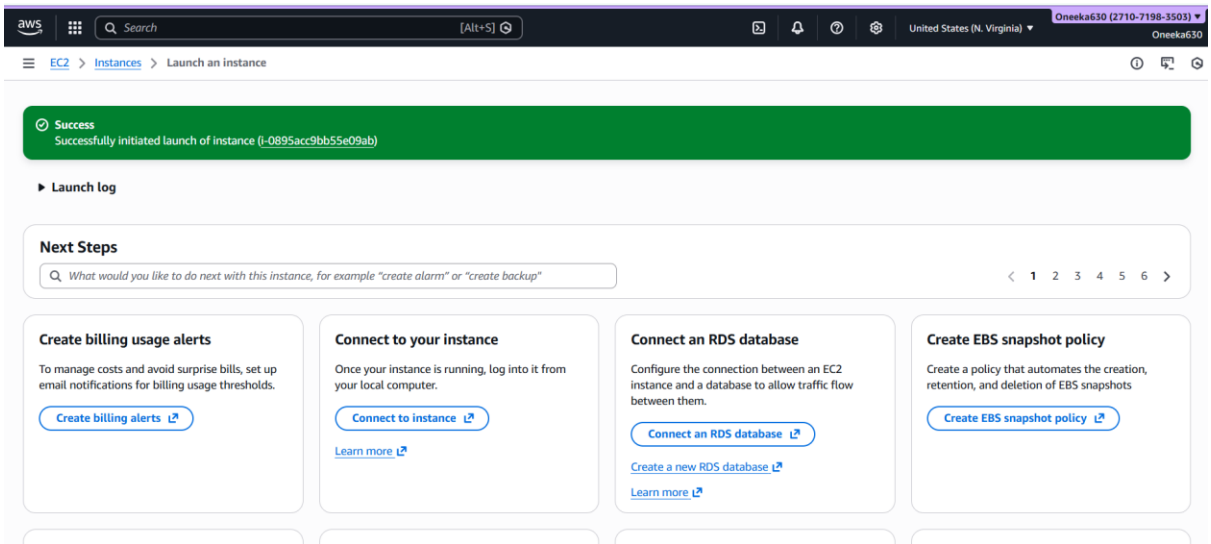



3) In **'Key pair (login)'**, click on '**Create new key pair'**, give '**key pair name'** and create it. Under '**Network Settings'** tick on all checkboxes. Then **Launch instance**.

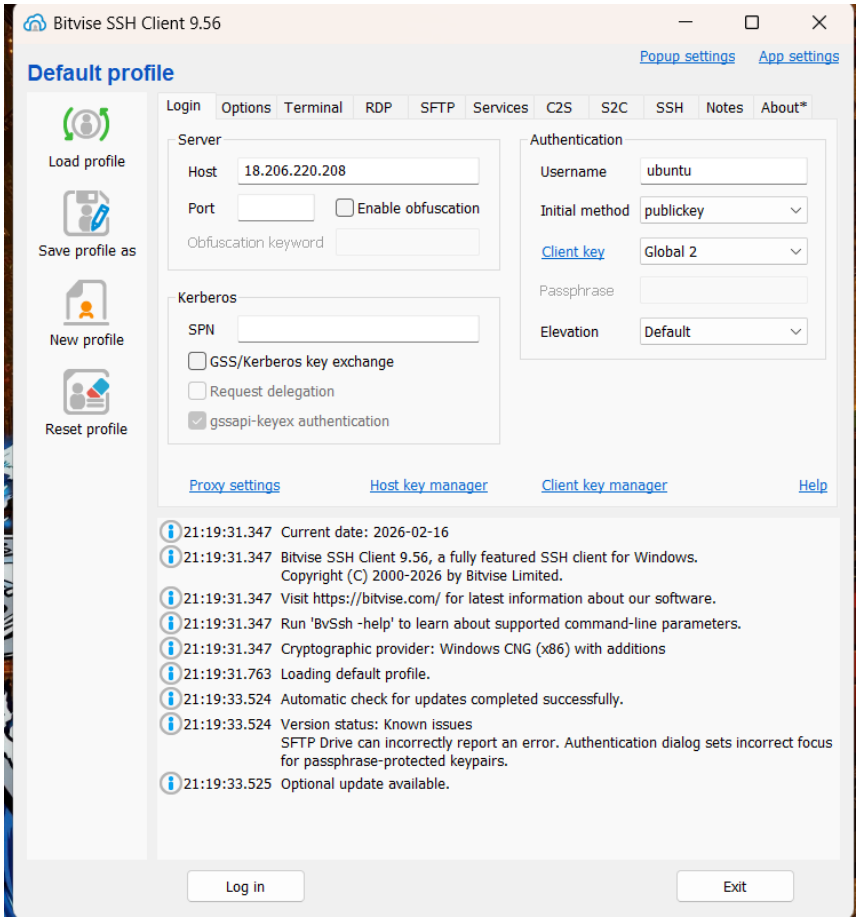


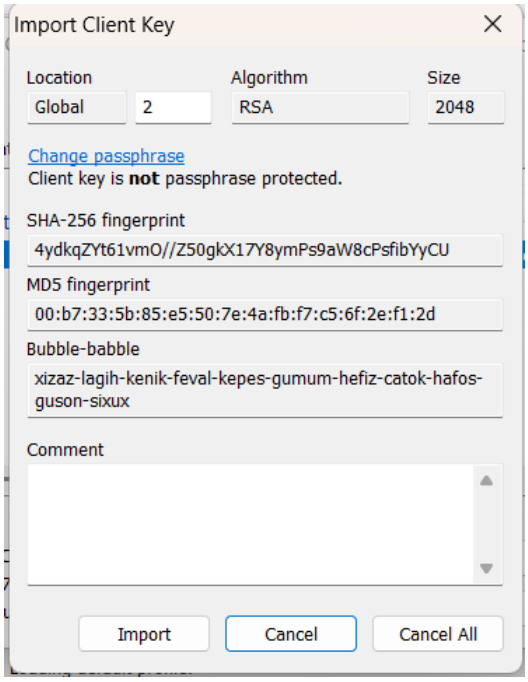

4) **'Oneeka001'** instance is created successfully.

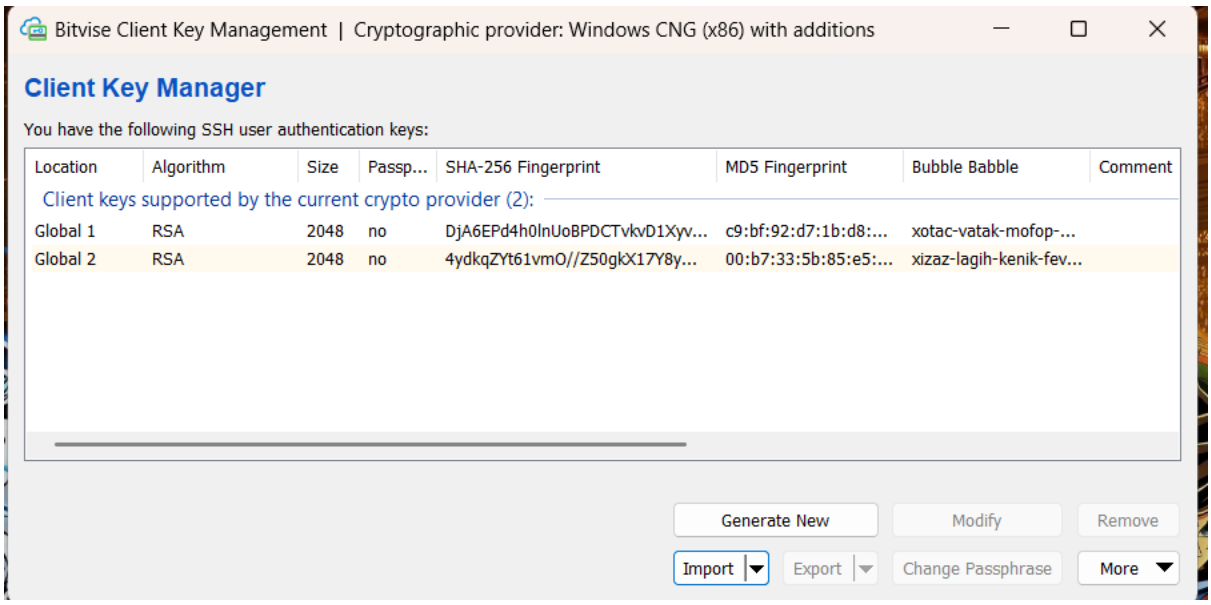



5) Click on 'Public IPV4 address' & copy it then open '**Bitvise SSH Client'**. In Bitvise SSH Client, paste the **'Public IPV4 address'** & click on '**Client key Manager'.**

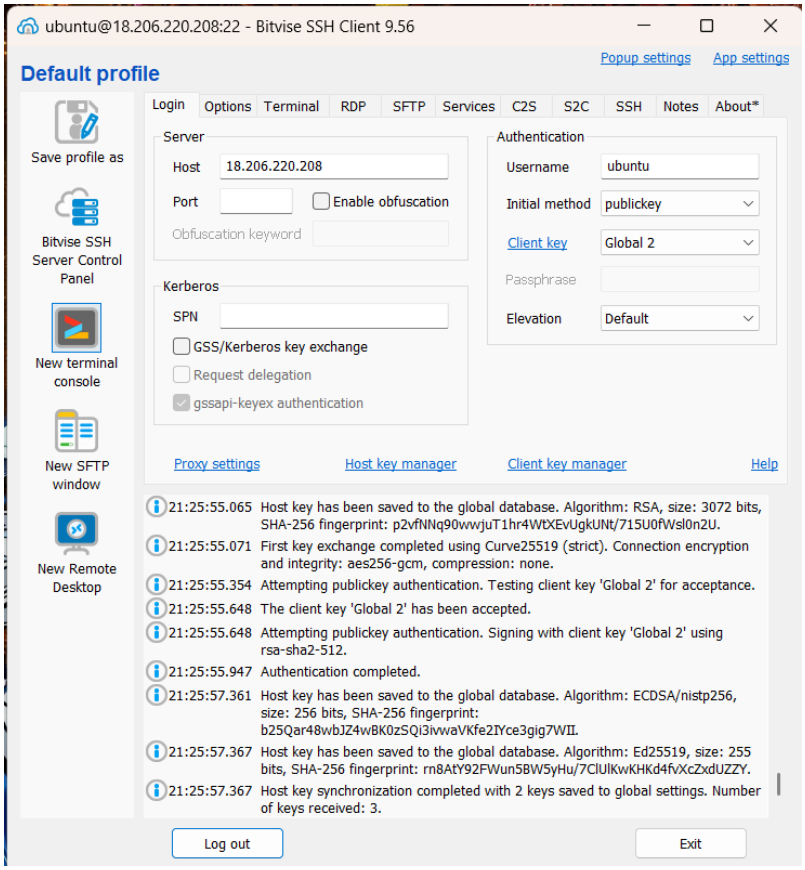



6)Under ‘**Client key manager’**, if there is any existing key then remove it and click on ‘**Import’**, select the created key ‘**key2.pem’** then again click on ‘**Import’** and then the key is successfully imported.






7)In '**Bitvise SSH Client',** under '**Default profile'** give the **'Username'** as **'ubuntu'** then select **Global2** from 'Client Key Manager' then click on '**Log in'** & '**Accept and save'.**




8)In 'Bitvise SSH client', under 'Default profile' click on '**New terminal console**' and there run **'sudo apt-get update'**, **'sudo apt-get upgrade'**.

```
ubuntu@ip-172-31-46-52:~$ sudo apt-get update && sudo apt-get upgrade
```

9)In terminal console, type the following commands. ->'**sudo apt-get install nginx', 'nginx -v', 'curl -sL https://deb.nodesource.com/setup_16.x | sudo -E bash –', 'sudo apt install nodejs'**

```
ubuntu@ip-172-31-46-52:~$ sudo apt-get install nginx
```

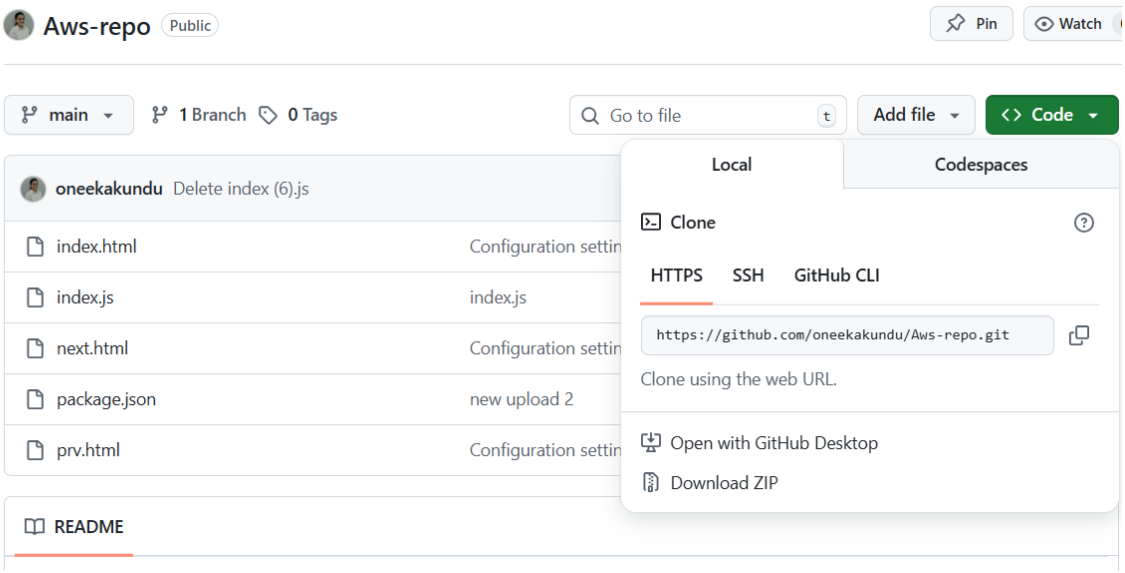
```
ubuntu@ip-172-31-46-52:~$ curl -sL https://deb.nodesource.com/setup_16.x|sudo -E bash -
```

```
ubuntu@ip-172-31-46-52:~$ sudo apt install nodejs
```

10)Now open **Github** and copy the **HTTPS** Link from the repository **'aws-repo'** and then run the commands

**'nodejs -v',**

**'git clone https://github.com/oneekakundu /Aws-repo.git'.**



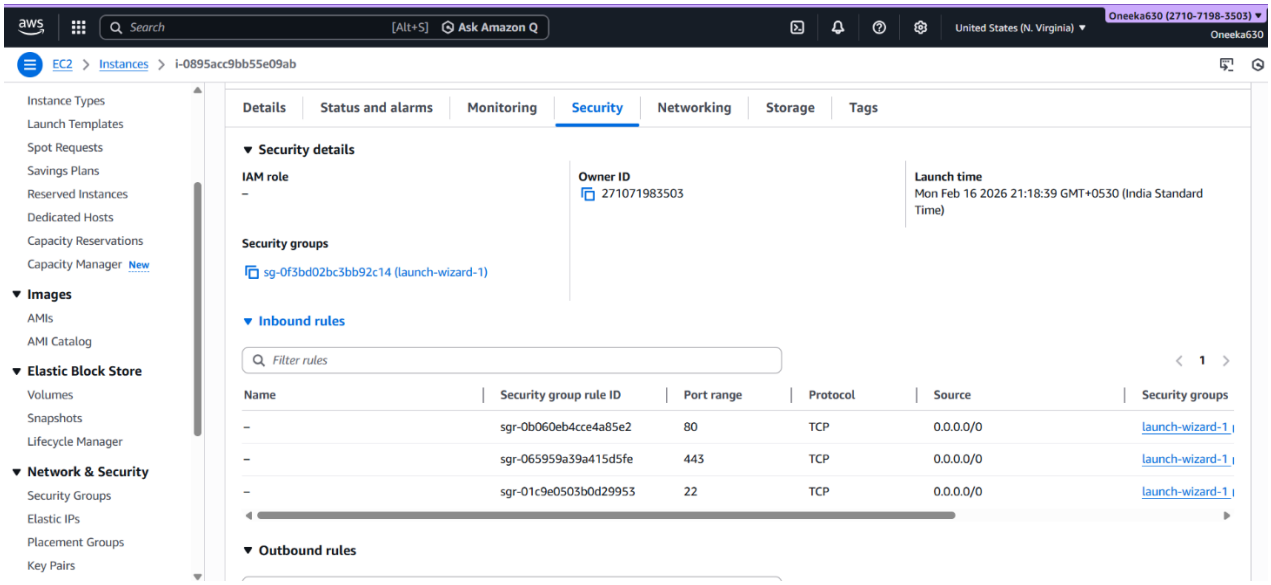

11) Now in the console, type also these following **commands** in sequential order.

```
ubuntu@ip-172-31-46-52:~$ git clone https://github.com/oneekakundu/Aws-repo.git
```

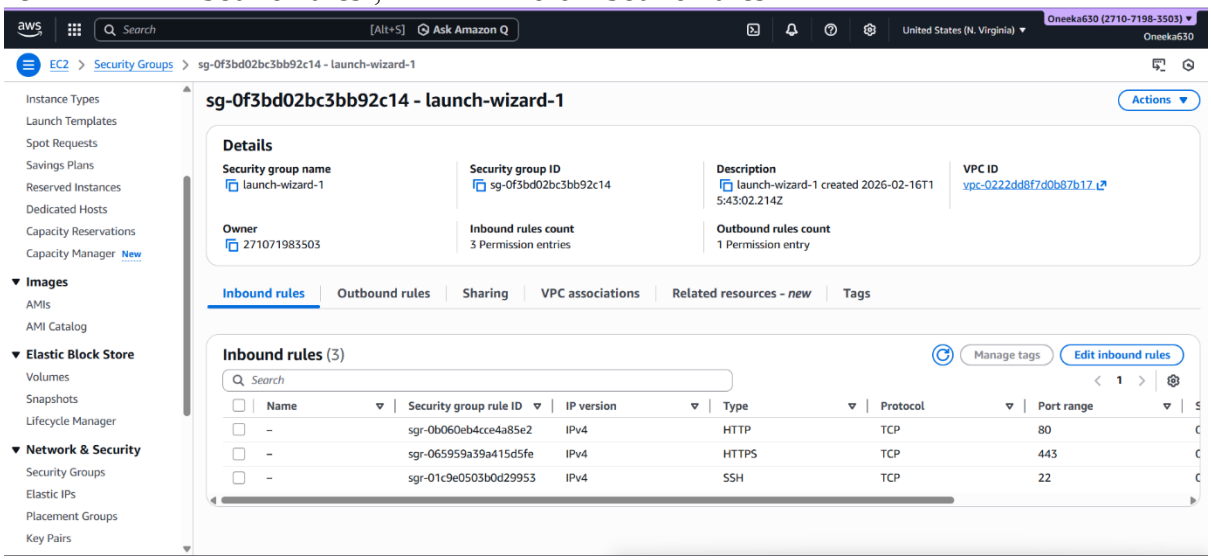
```
ubuntu@ip-172-31-46-52:~$ cd Aws-repo
ubuntu@ip-172-31-46-52:~/Aws-repo$ ls
index.html  index.js  next.html  package.json  prv.html
ubuntu@ip-172-31-46-52:~/Aws-repo$ npm install
```

```
ubuntu@ip-172-31-46-52:~/Aws-repo$ node index.js
Started server
```

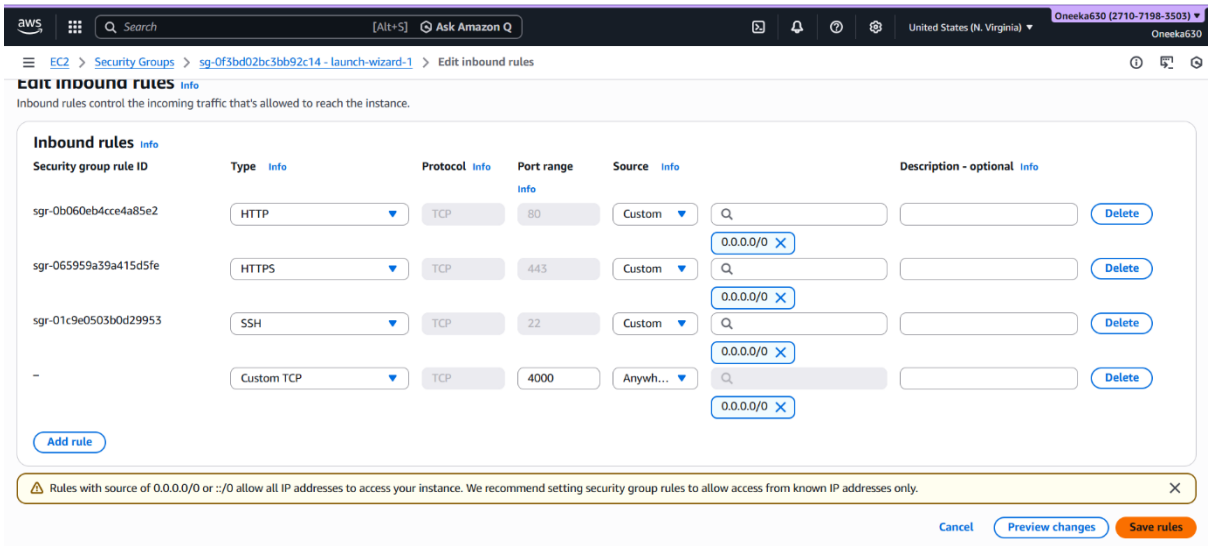
12. Now click on '**Instance ID**'-> Then go to '**Security**' and click on link under '**Security groups**' to get the edit option.



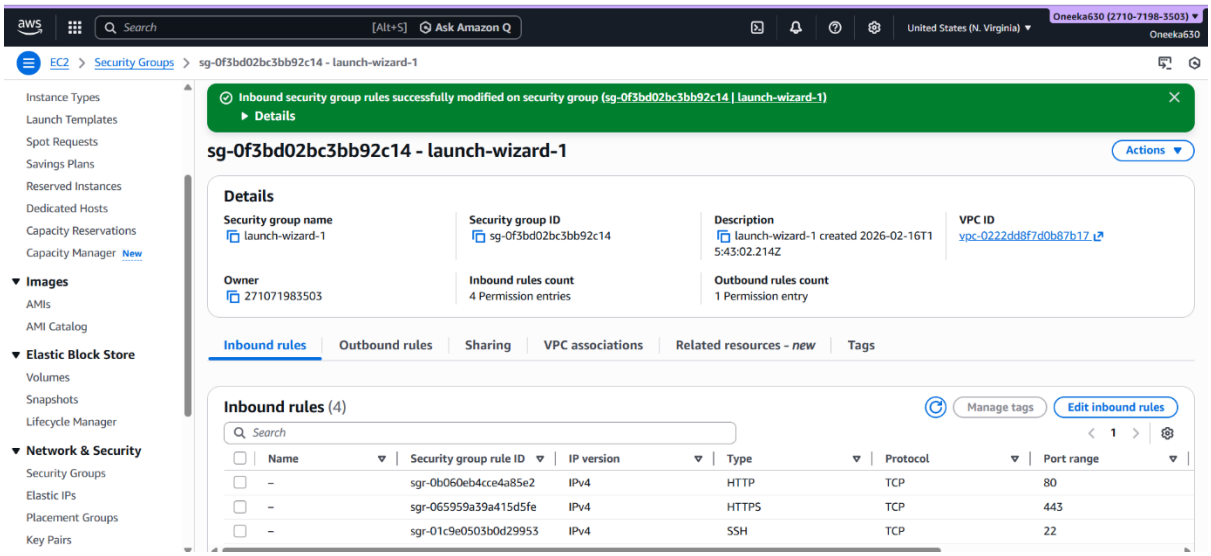

13. Then in '**Inbound rules**', click on '**Edit inbound rules**'.

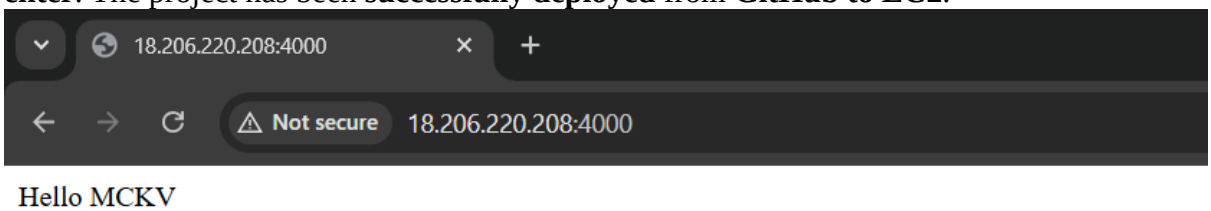



14. Click on '**Add rule**' -> Set the '**Port range**' to '**4000**', in '**Source**' set '**0.0.0.0/0**' & click on '**Save rules**'.




15. Rule has been **successfully saved**.




16. Now copy the '**Public IPv4 address**' & **paste** it on a new tab. Add **':4000'** at the end and then press **enter**. The project has been **successfully deployed** from **GitHub to EC2**.